

CP page 8 # 5b & 5c

$$5b) \quad y = [f(x)]^3$$

$$\frac{dy}{dx} = 3 [f(x)]^2 \cdot f'(x)$$

$$\left. \frac{dy}{dx} \right|_{x=\frac{1}{2}} = 3 \left[f\left(\frac{1}{2}\right) \right]^2 f'\left(\frac{1}{2}\right)$$

$$5c) \quad y = (1 + f(x))^2$$

$$\frac{dy}{dx} = 2(1 + f(x)) \cdot f'(x)$$

$$\left. \frac{dy}{dx} \right|_{x=\frac{1}{2}} = 2 \left(1 + f\left(\frac{1}{2}\right) \right) \cdot f'\left(\frac{1}{2}\right)$$

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$$\#6) \quad y = f(x^2 + 3x - 5)$$

$$\frac{dy}{dx} = f'(x^2 + 3x - 5) \cdot (2x + 3)$$

$$\frac{dy}{dx} \Big|_{x=1} = f'(1^2 + 3(1) - 5) (2(1) + 3)$$

$$= f'(-1)(5)$$

$$= 2(5)$$

$$= 10$$